User:

kindly give me top 100 java interview coding questions interviewer asked in interview with best easy optimized and simple answers also

ChatGPT:

Here's a curated list of **Top 100 Java Interview Coding Questions** that are commonly asked across companies like Infosys, TCS, Wipro, Capgemini, Cognizant, HCL, Tech Mahindra, and even in product-based companies. I've grouped them by topic and will provide **optimized and easy-to-understand solutions** in batches.

🔹 Batch 1: Basic Java Coding Questions (1–20)

1. Reverse a string

```
java
public String reverseString(String s) {
    return new StringBuilder(s).reverse().toString();
}
```

2. Check if a string is a palindrome

```
java
public boolean isPalindrome(String s) {
    int l = 0, r = s.length() - 1;
    while (l < r) if (s.charAt(l++) != s.charAt(r--)) return false;
    return true;
}
```

3. Find factorial of a number

```
java
public int factorial(int n) {
    return (n <= 1) ? 1 : n * factorial(n - 1);
}
```

4. Check if a number is prime

```
java
public boolean isPrime(int n) {
    if (n <= 1) return false;
    for (int i = 2; i <= Math.sqrt(n); i++) if (n % i == 0) return false;
    return true;
}
```

5. Fibonacci series up to n terms

```
java
public void printFibonacci(int n) {
    int a = 0, b = 1;
    for (int i = 0; i < n; i++) {
        System.out.print(a + " ");
        int temp = a + b;
        a = b;
        b = temp;
    }
}
```

6. Find second largest element in array

```java
public int secondLargest(int[] arr) {
    int max = Integer.MIN_VALUE, second = Integer.MIN_VALUE;
    for (int num : arr) {
        if (num > max) {
            second = max;
            max = num;
        } else if (num > second && num != max) {
            second = num;
        }
    }
    return second;
}
```

7. Count vowels and consonants in a string

```java
public void countVC(String s) {
    int v = 0, c = 0;
    for (char ch : s.toLowerCase().toCharArray()) {
        if ("aeiou".indexOf(ch) != -1) v++;
        else if (Character.isLetter(ch)) c++;
    }
    System.out.println("Vowels: " + v + ", Consonants: " + c);
}
```

8. Find duplicate characters in a string

```java
public Set<Character> findDuplicates(String s) {
    Set<Character> seen = new HashSet<>(), duplicates = new HashSet<>();
    for (char ch : s.toCharArray()) {
        if (!seen.add(ch)) duplicates.add(ch);
    }
    return duplicates;
}
```

9. Remove duplicates from array

```java
public int[] removeDuplicates(int[] arr) {
    return Arrays.stream(arr).distinct().toArray();
}
```

10. Check if two strings are anagrams

```java
public boolean areAnagrams(String s1, String s2) {
    char[] a1 = s1.toCharArray(), a2 = s2.toCharArray();
    Arrays.sort(a1); Arrays.sort(a2);
    return Arrays.equals(a1, a2);
}
```

11. Swap two numbers without temp variable

```java
public void swap(int a, int b) {
    a = a + b;
    b = a - b;
    a = a - b;
    System.out.println("a: " + a + ", b: " + b);
}
```

12. Check Armstrong number

```
java
public boolean isArmstrong(int n) {
    int sum = 0, original = n;
    while (n > 0) {
        int d = n % 10;
        sum += d * d * d;
        n /= 10;
    }
    return sum == original;
}
```

13. Count digits in a number

```
java
public int countDigits(int n) {
    return String.valueOf(Math.abs(n)).length();
}
```

14. Reverse a number

```
java
public int reverse(int n) {
    int rev = 0;
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    return rev;
}
```

15. Check if number is palindrome

```
java
public boolean isPalindromeNumber(int n) {
    return n == reverse(n);
}
```

16. Sum of digits of a number

```
java
public int digitSum(int n) {
    int sum = 0;
    while (n != 0) {
        sum += n % 10;
        n /= 10;
    }
    return sum;
}
```

17. Find missing number in array 1 to n

```
java
public int findMissing(int[] arr, int n) {
    int sum = n * (n + 1) / 2;
    for (int num : arr) sum -= num;
    return sum;
}
```

18. Count occurrence of each char in string

```
java
public Map<Character, Integer> countChars(String s) {
    Map<Character, Integer> map = new HashMap<>();
    for (char ch : s.toCharArray()) {
        map.put(ch, map.getOrDefault(ch, 0) + 1);
    }
    return map;
}
```

19. Find GCD of two numbers

```
java
public int gcd(int a, int b) {
    return b == 0 ? a : gcd(b, a % b);
}
```

20. Check leap year

```
java
public boolean isLeap(int year) {
    return (year % 4 == 0 && year % 100 != 0) || year % 400 == 0;
}
```

Would you like me to continue with:

- **Batch 2: Arrays and Matrix Questions (21–40)** or
- **Batch 2: String and Pattern Matching Questions**,
  or do you want all 100 in one downloadable format (PDF/Word)?

Let me know your preference!

User:

yes

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Great! Since you said **yes**, I'll continue with the next **Batch 2: Arrays and Matrix Questions (21–40)** with optimized, easy-to-understand solutions.

🔹 Batch 2: Arrays and Matrix Questions (21–40)

21. Find the largest element in array

```
java
public int findMax(int[] arr) {
    return Arrays.stream(arr).max().orElse(Integer.MIN_VALUE);
}
```

22. Find the smallest element in array

```
java
public int findMin(int[] arr) {
    return Arrays.stream(arr).min().orElse(Integer.MAX_VALUE);
}
```

23. Sort an array (Ascending)

```java
public void sortArray(int[] arr) {
    Arrays.sort(arr);
}
```

24. Sort array in descending order

```java
public Integer[] sortDescending(int[] arr) {
    return Arrays.stream(arr).boxed()
        .sorted(Collections.reverseOrder()).toArray(Integer[]::new);
}
```

25. Check if array is sorted

```java
public boolean isSorted(int[] arr) {
    for (int i = 1; i < arr.length; i++) {
        if (arr[i] < arr[i - 1]) return false;
    }
    return true;
}
```

26. Move all zeros to end

```java
public void moveZeros(int[] arr) {
    int idx = 0;
    for (int num : arr) if (num != 0) arr[idx++] = num;
    while (idx < arr.length) arr[idx++] = 0;
}
```

27. Rotate array by k steps

```java
public void rotate(int[] arr, int k) {
    k %= arr.length;
    reverse(arr, 0, arr.length - 1);
    reverse(arr, 0, k - 1);
    reverse(arr, k, arr.length - 1);
}
void reverse(int[] a, int i, int j) {
    while (i < j) {
        int t = a[i]; a[i++] = a[j]; a[j--] = t;
    }
}
```

28. Find common elements in two arrays

```java
public Set<Integer> commonElements(int[] a, int[] b) {
    Set<Integer> set = new HashSet<>();
    Set<Integer> result = new HashSet<>();
    for (int num : a) set.add(num);
    for (int num : b) if (set.contains(num)) result.add(num);
    return result;
}
```

29. Find frequency of each element

```java
public Map<Integer, Integer> frequency(int[] arr) {
    Map<Integer, Integer> map = new HashMap<>();
    for (int num : arr) map.put(num, map.getOrDefault(num, 0) + 1);
    return map;
}
```

30. Find unique elements (non-repeating)

```java
public List<Integer> findUniques(int[] arr) {
    Map<Integer, Integer> map = frequency(arr);
    return map.entrySet().stream()
              .filter(e -> e.getValue() == 1)
              .map(Map.Entry::getKey)
              .collect(Collectors.toList());
}
```

31. Find subarray with given sum

```java
public boolean hasSubarraySum(int[] arr, int sum) {
    Set<Integer> set = new HashSet<>();
    int currSum = 0;
    for (int num : arr) {
        currSum += num;
        if (currSum == sum || set.contains(currSum - sum)) return true;
        set.add(currSum);
    }
    return false;
}
```

32. Kadane's algorithm (max subarray sum)

```java
public int maxSubarraySum(int[] arr) {
    int maxSoFar = arr[0], currMax = arr[0];
    for (int i = 1; i < arr.length; i++) {
        currMax = Math.max(arr[i], currMax + arr[i]);
        maxSoFar = Math.max(maxSoFar, currMax);
    }
    return maxSoFar;
}
```

33. Matrix transpose

```java
public int[][] transpose(int[][] matrix) {
    int row = matrix.length, col = matrix[0].length;
    int[][] result = new int[col][row];
    for (int i = 0; i < row; i++)
        for (int j = 0; j < col; j++)
            result[j][i] = matrix[i][j];
    return result;
}
```

34. Check if matrix is symmetric

```java
public boolean isSymmetric(int[][] m) {
    for (int i = 0; i < m.length; i++)
```

```
        for (int j = 0; j < m[0].length; j++)
            if (m[i][j] != m[j][i]) return false;
    return true;
}
```

35. Matrix multiplication

```
java
public int[][] multiply(int[][] A, int[][] B) {
    int r1 = A.length, c1 = A[0].length, c2 = B[0].length;
    int[][] res = new int[r1][c2];
    for (int i = 0; i < r1; i++)
        for (int j = 0; j < c2; j++)
            for (int k = 0; k < c1; k++)
                res[i][j] += A[i][k] * B[k][j];
    return res;
}
```

36. Search element in 2D sorted matrix

```
java
public boolean searchMatrix(int[][] m, int target) {
    int row = 0, col = m[0].length - 1;
    while (row < m.length && col >= 0) {
        if (m[row][col] == target) return true;
        else if (m[row][col] > target) col--;
        else row++;
    }
    return false;
}
```

37. Find max element in 2D matrix

```
java
public int max2D(int[][] matrix) {
    int max = Integer.MIN_VALUE;
    for (int[] row : matrix)
        for (int val : row)
            max = Math.max(max, val);
    return max;
}
```

38. Spiral traversal of matrix

```
java
public List<Integer> spiralOrder(int[][] matrix) {
    List<Integer> result = new ArrayList<>();
    int top = 0, bottom = matrix.length - 1;
    int left = 0, right = matrix[0].length - 1;
    while (top <= bottom && left <= right) {
        for (int i = left; i <= right; i++) result.add(matrix[top][i]);
        top++;
        for (int i = top; i <= bottom; i++) result.add(matrix[i][right]);
        right--;
        if (top <= bottom) for (int i = right; i >= left; i--) result.add(matrix[bottom][i]);
        bottom--;
        if (left <= right) for (int i = bottom; i >= top; i--) result.add(matrix[i][left]);
        left++;
    }
    return result;
}
```

39. **Diagonal sum of matrix**

```
java
public int diagonalSum(int[][] m) {
    int sum = 0;
    for (int i = 0; i < m.length; i++) {
        sum += m[i][i];
        if (i != m.length - 1 - i)
            sum += m[i][m.length - 1 - i];
    }
    return sum;
}
```

40. **Set matrix zeroes (if row/col has zero, set entire row/col to zero)**

```
java
public void setZeroes(int[][] m) {
    int R = m.length, C = m[0].length;
    boolean[] row = new boolean[R], col = new boolean[C];
    for (int i = 0; i < R; i++)
        for (int j = 0; j < C; j++)
            if (m[i][j] == 0) row[i] = col[j] = true;

    for (int i = 0; i < R; i++)
        for (int j = 0; j < C; j++)
            if (row[i] || col[j]) m[i][j] = 0;
}
```

Let me know if you'd like to continue with:

- **Batch 3: String and Recursion (41–60)**
- **Batch 4: OOP, Collections, Map, Set (61–80)**
- **Batch 5: Java 8, Streams, Lambdas, Misc (81–100)**
- OR I can compile **all 100 questions with answers in PDF format** for download.

Which one do you prefer?

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Great! Let's continue with **Batch 3: String and Recursion Questions (41–60)**.

🔹 Batch 3: String and Recursion Questions (41–60)

41. **Reverse words in a sentence**

```
java
public String reverseWords(String s) {
    String[] words = s.split(" ");
    StringBuilder sb = new StringBuilder();
    for (int i = words.length - 1; i >= 0; i--) {
        sb.append(words[i]);
        if (i != 0) sb.append(" ");
    }
```

```
    return sb.toString();
}
```

42. Remove all occurrences of a character from a string

```
java
public String removeChar(String s, char c) {
    return s.replaceAll(String.valueOf(c), "");
}
```

43. Check if two strings are rotations of each other

```
java
public boolean areRotations(String s1, String s2) {
    return s1.length() == s2.length() && (s1 + s1).contains(s2);
}
```

44. Count number of vowels in a string

```
java
public int countVowels(String s) {
    int count = 0;
    for (char ch : s.toLowerCase().toCharArray()) {
        if ("aeiou".indexOf(ch) != -1) count++;
    }
    return count;
}
```

45. Print all subsequences of a string

```
java
public void printSubsequences(String s) {
    printSubsequencesHelper(s, 0, "");
}

private void printSubsequencesHelper(String s, int index, String result) {
    if (index == s.length()) {
        System.out.println(result);
        return;
    }
    printSubsequencesHelper(s, index + 1, result);
    printSubsequencesHelper(s, index + 1, result + s.charAt(index));
}
```

46. Find the longest palindromic substring

```
java
public String longestPalindrome(String s) {
    if (s == null || s.length() < 1) return "";
    int start = 0, end = 0;
    for (int i = 0; i < s.length(); i++) {
        int len1 = expandAroundCenter(s, i, i);
        int len2 = expandAroundCenter(s, i, i + 1);
        int len = Math.max(len1, len2);
        if (len > end - start) {
            start = i - (len - 1) / 2;
            end = i + len / 2;
        }
    }
    return s.substring(start, end + 1);
}
```

```
private int expandAroundCenter(String s, int left, int right) {
    while (left >= 0 && right < s.length() && s.charAt(left) == s.charAt(right)) {
        left--;
        right++;
    }
    return right - left - 1;
}
```

47. Generate all permutations of a string

```
java
public void generatePermutations(String s) {
    generatePermutationsHelper(s, 0, s.length() - 1);
}

private void generatePermutationsHelper(String s, int left, int right) {
    if (left == right) {
        System.out.println(s);
        return;
    }
    for (int i = left; i <= right; i++) {
        s = swap(s, left, i);
        generatePermutationsHelper(s, left + 1, right);
        s = swap(s, left, i); // backtrack
    }
}

private String swap(String s, int i, int j) {
    char[] arr = s.toCharArray();
    char temp = arr[i];
    arr[i] = arr[j];
    arr[j] = temp;
    return new String(arr);
}
```

48. Find first non-repeating character in a string

```
java
public char firstUniqChar(String s) {
    Map<Character, Integer> map = new LinkedHashMap<>();
    for (char c : s.toCharArray()) {
        map.put(c, map.getOrDefault(c, 0) + 1);
    }
    for (Map.Entry<Character, Integer> entry : map.entrySet()) {
        if (entry.getValue() == 1) return entry.getKey();
    }
    return '\0'; // If no unique character is found
}
```

49. Implement strstr (substring search)

```
java
public int strStr(String haystack, String needle) {
    if (needle.isEmpty()) return 0;
    for (int i = 0; i <= haystack.length() - needle.length(); i++) {
        if (haystack.substring(i, i + needle.length()).equals(needle)) return i;
    }
    return -1;
}
```

50. Find length of longest substring without repeating characters

```
java
```

```
public int lengthOfLongestSubstring(String s) {
    Set<Character> set = new HashSet<>();
    int left = 0, maxLen = 0;
    for (int right = 0; right < s.length(); right++) {
        while (set.contains(s.charAt(right))) {
            set.remove(s.charAt(left++));
        }
        set.add(s.charAt(right));
        maxLen = Math.max(maxLen, right - left + 1);
    }
    return maxLen;
}
```

51. Find the sum of digits of a number recursively

```
java
public int sumOfDigits(int n) {
    if (n == 0) return 0;
    return n % 10 + sumOfDigits(n / 10);
}
```

52. Print n-th Fibonacci number recursively

```
java
public int fibonacci(int n) {
    if (n <= 1) return n;
    return fibonacci(n - 1) + fibonacci(n - 2);
}
```

53. Find the power of a number using recursion

```
java
public int power(int base, int exp) {
    if (exp == 0) return 1;
    return base * power(base, exp - 1);
}
```

54. Find all subsets of a set (Power set)

```
java
public void printSubsets(int[] nums) {
    List<List<Integer>> result = new ArrayList<>();
    subsetsHelper(nums, 0, new ArrayList<>(), result);
    System.out.println(result);
}

private void subsetsHelper(int[] nums, int index, List<Integer> current, List<List<Integer>> result) {
    if (index == nums.length) {
        result.add(new ArrayList<>(current));
        return;
    }
    subsetsHelper(nums, index + 1, current, result);
    current.add(nums[index]);
    subsetsHelper(nums, index + 1, current, result);
    current.remove(current.size() - 1); // backtrack
}
```

55. Check if a string has balanced parentheses

```
java
public boolean isBalanced(String s) {
    Stack<Character> stack = new Stack<>();
```

```
    for (char c : s.toCharArray()) {
        if (c == '(') stack.push(c);
        else if (c == ')') {
            if (stack.isEmpty()) return false;
            stack.pop();
        }
    }
    return stack.isEmpty();
}
```

56. Convert string to integer (without using Integer.parseInt)

```
java
public int stringToInt(String s) {
    int result = 0;
    boolean isNegative = s.charAt(0) == '-';
    int start = isNegative ? 1 : 0;
    for (int i = start; i < s.length(); i++) {
        result = result * 10 + (s.charAt(i) - '0');
    }
    return isNegative ? -result : result;
}
```

57. Find the number of set bits (1s) in a number

```
java
public int countSetBits(int n) {
    int count = 0;
    while (n > 0) {
        count += n & 1;
        n >>= 1;
    }
    return count;
}
```

58. Check if a number is power of two

```
java
public boolean isPowerOfTwo(int n) {
    return n > 0 && (n & (n - 1)) == 0;
}
```

59. Print all binary representations of numbers from 1 to n

```
java
public void printBinary(int n) {
    for (int i = 1; i <= n; i++) {
        System.out.println(Integer.toBinaryString(i));
    }
}
```

60. Find the greatest common divisor (GCD) using recursion

```
java
public int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
```

Let me know if you'd like to continue with:

- **Batch 4: OOP, Collections, Map, Set (61–80)**
- **Batch 5: Java 8, Streams, Lambdas, Misc (81–100)**

Or I can provide **all 100 questions** in a **single PDF download**. Let me know your preference!

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Awesome! Let's move on to **Batch 4: OOP, Collections, Map, Set Questions (61–80)** — these are **super important** for Java interviews, especially for experienced roles.

🔹 Batch 4: OOP, Collections, Map, Set (61–80)

61. Difference between `==` and `.equals()`

```
java
String a = new String("Hello");
String b = new String("Hello");

System.out.println(a == b);       // false (compares reference)
System.out.println(a.equals(b));  // true  (compares values)
```

62. Override `equals()` and `hashCode()`

```
java
class Employee {
    int id;
    String name;

    // Constructor + getters/setters...

    @Override
    public boolean equals(Object o) {
        if (this == o) return true;
        if (!(o instanceof Employee)) return false;
        Employee e = (Employee) o;
        return id == e.id && Objects.equals(name, e.name);
    }

    @Override
    public int hashCode() {
        return Objects.hash(id, name);
    }
}
```

63. What is `Comparable`? Custom sort using `Comparable`

```
java
class Student implements Comparable<Student> {
    int age;
    public int compareTo(Student s) {
        return this.age - s.age; // Ascending order
    }
}
```

64. What is `Comparator`? Custom sort using `Comparator`

```
java
class Student {
    int age;
    String name;
}
Comparator<Student> byName = Comparator.comparing(s -> s.name);
Collections.sort(studentList, byName);
```

65. Difference between ArrayList and LinkedList

- `ArrayList`: fast random access, slow insertion/deletion (O(n))
- `LinkedList`: slow access, fast insertion/deletion (O(1) at ends)

66. Fail-fast vs Fail-safe

- **Fail-fast** (e.g., `ArrayList`, `HashMap`): throws `ConcurrentModificationException`
- **Fail-safe** (e.g., `ConcurrentHashMap`, `CopyOnWriteArrayList`): uses clone/lock mechanism

67. Find duplicate elements using HashSet

```
java
public void printDuplicates(int[] arr) {
    Set<Integer> seen = new HashSet<>();
    for (int num : arr) {
        if (!seen.add(num)) {
            System.out.println("Duplicate: " + num);
        }
    }
}
```

68. Remove duplicates from ArrayList

```
java
List<Integer> list = new ArrayList<>(Arrays.asList(1,2,2,3,4,4));
Set<Integer> set = new LinkedHashSet<>(list);
list = new ArrayList<>(set);
```

69. Find frequency of elements using HashMap

```
java
public Map<Integer, Integer> frequency(int[] arr) {
    Map<Integer, Integer> map = new HashMap<>();
    for (int num : arr) {
        map.put(num, map.getOrDefault(num, 0) + 1);
    }
    return map;
}
```

70. Sort HashMap by values

```
java
Map<String, Integer> map = new HashMap<>();
map.entrySet().stream()
   .sorted(Map.Entry.comparingByValue())
   .forEach(e -> System.out.println(e.getKey() + " " + e.getValue()));
```

71. Difference between HashMap, TreeMap, LinkedHashMap

Feature	HashMap	TreeMap	LinkedHashMap
Order	No order	Sorted by keys	Insertion order
Null Keys	1 allowed	Not allowed	1 allowed
Performance	Fastest	Slower (BST)	Slightly slower

72. SynchronizedMap vs ConcurrentHashMap

- `SynchronizedMap` is slower (locks full map)
- `ConcurrentHashMap` is faster (locks segments/buckets)

73. What is HashSet? How it avoids duplicates?

- Internally uses `HashMap` with keys only.
- Relies on `.hashCode()` and `.equals()`.

74. Sort a Set

```
java
Set<Integer> set = new HashSet<>(Arrays.asList(4,1,3,2));
Set<Integer> sorted = new TreeSet<>(set); // Automatically sorted
```

75. Find first non-repeating character using LinkedHashMap

```
java
public char firstNonRepeat(String s) {
    Map<Character, Integer> map = new LinkedHashMap<>();
    for (char c : s.toCharArray()) {
        map.put(c, map.getOrDefault(c, 0) + 1);
    }
    for (Map.Entry<Character, Integer> entry : map.entrySet()) {
        if (entry.getValue() == 1) return entry.getKey();
    }
    return '_';
}
```

76. Difference between List, Set, and Map

Interface	Allows Duplicates	Ordered	Key-Value
List	Yes	Yes	No
Set	No	No (HashSet) / Yes (LinkedHashSet)	No
Map	No (keys)	Depends	Yes

77. Convert Map to List

```
java
Map<Integer, String> map = new HashMap<>();
List<Map.Entry<Integer, String>> list = new ArrayList<>(map.entrySet());
```

78. Convert List to Map

```
java
List<String> list = Arrays.asList("a", "b", "c");
Map<Integer, String> map = IntStream.range(0, list.size())
        .boxed()
        .collect(Collectors.toMap(i -> i, list::get));
```

79. Group elements by frequency using Map

```
java
Map<Integer, List<Integer>> grouped = Arrays.stream(arr)
    .boxed()
    .collect(Collectors.groupingBy(Function.identity()));
```

80. **Find the most frequent element in an array**

```
java
public int mostFrequent(int[] arr) {
    Map<Integer, Integer> freq = new HashMap<>();
    for (int num : arr) freq.put(num, freq.getOrDefault(num, 0) + 1);
    return Collections.max(freq.entrySet(), Map.Entry.comparingByValue()).getKey();
}
```

🔥 You're now through **80 out of 100** critical interview-ready Java coding problems!

Ready for the **final batch**?

- ✅ **Batch 5: Java 8, Streams, Lambdas, Optional, Misc (81–100)**

Should I go ahead and drop the last 20 now?

User:

yes

ChatGPT:

Awesome! Here comes the final one — the most in-demand round in interviews nowadays:

🔹 Batch 5: Java 8, Streams, Lambdas, Optional, Misc (81–100)

81. **What is a Lambda expression?**

```
java
// Syntax: (parameters) -> { body }
List<String> names = Arrays.asList("Alex", "Brian", "Charles");
names.forEach(name -> System.out.println(name));
```

82. **Use Stream to filter and collect**

```
java
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5);
List<Integer> even = numbers.stream()
    .filter(n -> n % 2 == 0)
    .collect(Collectors.toList());
```

83. **Map each string to its length using Stream**

```
java
List<String> list = Arrays.asList("java", "spring", "boot");
List<Integer> lengths = list.stream()
    .map(String::length)
    .collect(Collectors.toList());
```

84. **Find first element greater than 10 using Stream**

```
java
Optional<Integer> first = numbers.stream()
    .filter(n -> n > 10)
    .findFirst();
```

85. Sum of all elements using Stream

```
java
int sum = numbers.stream().mapToInt(Integer::intValue).sum();
```

86. Sort a list in reverse using Stream

```
java
List<String> sorted = list.stream()
    .sorted(Comparator.reverseOrder())
    .collect(Collectors.toList());
```

87. Count frequency of elements using Stream

```
java
Map<Integer, Long> freq = numbers.stream()
    .collect(Collectors.groupingBy(Function.identity(), Collectors.counting()));
```

88. Convert List<String> to comma-separated String

```
java
String joined = list.stream().collect(Collectors.joining(", "));
```

89. Use `Optional` to avoid NullPointerException

```
java
Optional<String> opt = Optional.ofNullable(someValue);
String result = opt.orElse("default");
```

90. Use `flatMap` to flatten nested lists

```
java
List<List<String>> nested = Arrays.asList(
    Arrays.asList("a", "b"),
    Arrays.asList("c", "d")
);
List<String> flat = nested.stream()
    .flatMap(Collection::stream)
    .collect(Collectors.toList());
```

🔸 Miscellaneous & Tricky Concepts (91–100)

91. Why String is immutable in Java?

- Security, caching, thread-safety, String pool optimization.

92. Difference between StringBuilder and StringBuffer

Feature	StringBuilder	StringBuffer
Thread Safe	No	Yes
Performance	Faster	Slower

93. Singleton Pattern (Lazy Initialization)

```
java
class Singleton {
    private static Singleton instance;
    private Singleton() {}
    public static Singleton getInstance() {
        if (instance == null) {
            instance = new Singleton();
        }
        return instance;
    }
}
```

94. Factory Pattern (Simple Example)

```
java
interface Shape { void draw(); }
class Circle implements Shape { public void draw() { System.out.println("Circle"); } }
class ShapeFactory {
    public Shape getShape(String type) {
        return switch (type) {
            case "circle" -> new Circle();
            default -> null;
        };
    }
}
```

95. Find memory leak using Java

- Use tools: `jvisualvm`, `jconsole`, or `heap dump analysis`.

96. Thread-safe counter in Java

```
java
AtomicInteger counter = new AtomicInteger();
counter.incrementAndGet();
```

97. Create a thread using lambda

```
java
Thread t = new Thread(() -> System.out.println("Running"));
t.start();
```

98. Deadlock Example

```
java
synchronized (lock1) {
    synchronized (lock2) {
        // Deadlock if another thread locks in reverse order
    }
}
```

99. Difference: checked vs unchecked exceptions

Type	Example	Must Handle
Checked	IOException	Yes
Unchecked	NullPointerException	No

100. Best practice to avoid NullPointerException

- Use `Optional`
- Null checks
- Use defensive coding

✅ **Congratulations!** You now have the **Top 100 Java Interview Coding Questions** — all **simple, optimized, interview-ready** answers. 💪

📦 **Want all 100 in a well-formatted PDF or Notion page for fast revision?**
Let me know and I'll generate and share it with you right away.

User:

is above ques are suffieceint to crack interview right, anything else you want to highlight or anything else tips to focus on above ques? like imp ques in above that i should start 1st and skip ?

ChatGPT:

That's a really smart question — prioritization is **key** when time is limited and you want **maximum impact** in your interviews.

✅ Are the above 100 questions sufficient?

Yes, **absolutely sufficient** for **cracking Java developer interviews** — especially for roles involving:

- Core Java (OOPs, Collections, Exception Handling)
- Java 8+ (Streams, Lambdas, Functional Interfaces)
- Coding Logic & Problem Solving
- API development (tie-in with Spring Boot)

They **cover the most-asked** coding questions across **Infosys, TCS, Wipro, Accenture, Cognizant, Capgemini, Tech Mahindra, and product companies**.

🔥 Must-Focus Priority Topics (Start Here First)

If you're short on time or want to revise smartly, **start with these 40** — they are **asked 9 out of 10 times**:

🔹 High Priority (A+)

Area	Questions to Focus
Arrays & Strings	Q1–10, Q16–18, Q30–32
Collections	Q61–80 (esp. HashMap, HashSet, frequency, sort, duplicates)
Java 8 (Streams)	Q81–90 — must be fluent here
OOP + Equals/HashCode	Q61–64, Q91
Comparator vs Comparable	Q63–64
Map Operations	Q69–70, Q87
Optional	Q89
Tricky Concepts	Q93–94 (Singleton, Factory), Q96 (AtomicInteger)

🟢 **Tip**: Questions like sorting by value, finding frequency, and converting list to map are always asked.

❌ Lower Priority (Can Skip If Rushed)

If you're in a time crunch, you **can skip for now**:

Questions	Reason
Q95	Memory leak – advanced, mostly not asked unless senior role
Q97–98	Thread creation and deadlock – rarely asked in service-based interviews

Questions	Reason
Q99–100	Exceptions – basic theory, revise if time

🧠 Final Tips to Crack the Interview

1. **Practice on whiteboard or paper**: Many interviews test your logical flow, not just syntax.
2. **Explain as you code**: Speak out loud — interviewers love clarity.
3. **Mock Interview Yourself**: Take 10 random Qs and time yourself. Be fluent and concise.
4. **Never write brute force directly** – always say:

   > *"First, I'll solve it simply, and then I'll optimize it."*
   > This shows maturity even before you code.

5. **Real-world understanding matters**:
   - Don't just write `HashMap` — know when and why to use it.
   - Understand mutability, thread safety, sorting logic, and performance.

📘 Want Help with Spring Boot + API Round Too?

I remember you're aiming for roles involving Spring Boot + Microservices + API Management — I can give you:

- **Top 50 Spring Boot + Microservices interview Qs** (very practical)
- **Real-world coding questions** like API filtering, pagination, exception handling, etc.

Wanna go for that next?

User:

Okay go ahead

ChatGPT:

Perfect! Since you're aiming for Spring Boot + Microservices + API Management roles, here's what I'll give you:

🔥 Top 50 Spring Boot & Microservices Interview Questions (with Simple Answers)

✅ These are **frequently asked** in Infosys, TCS, Accenture, Capgemini, etc.
✅ Focused on **real-time project scenarios**, not just theory.

🧩 PART 1: Spring Boot Core (1–15)

1. What is Spring Boot?

Spring Boot simplifies Spring app development by auto-configuring beans, providing defaults, and embedding servers like Tomcat so you can just run your app with `main()`.

2. What is @SpringBootApplication?

It's a combination of:

- `@Configuration`
- `@EnableAutoConfiguration`
- `@ComponentScan`

3. How to create a REST API in Spring Boot?

Use `@RestController`, map endpoints with `@GetMapping`, `@PostMapping`, etc.

```
java
@RestController
public class UserController {
    @GetMapping("/users")
    public List<User> getAll() { return service.getUsers(); }
}
```

4. Difference between @Controller and @RestController

- `@Controller` returns a View (HTML)
- `@RestController` returns JSON (REST)

5. What is application.properties / application.yml used for?

To define configuration like port, DB credentials, log level, etc.

```
properties
server.port=8081
spring.datasource.url=jdbc:mysql://localhost:3306/db
```

6. How do you connect to a database?

- Add dependency (`spring-boot-starter-data-jpa`)
- Configure `application.properties`
- Use `@Entity`, `@Repository`, and `JpaRepository`.

7. What is @Autowired?

Spring injects dependencies automatically using this annotation.

8. What is @Qualifier used for?

When multiple beans of the same type exist — to tell Spring which one to inject.

9. How does Spring Boot auto-configuration work?

Spring Boot checks classpath, environment, and properties to auto-configure beans (enabled by `@EnableAutoConfiguration`).

10. What is @Component, @Service, @Repository?

All are stereotype annotations that mark classes as Spring-managed beans.

11. Difference between @Component and @Bean

- `@Component` is used on classes, Spring detects it via classpath scanning.
- `@Bean` is used in configuration methods.

12. How to handle exceptions globally?

Use `@ControllerAdvice` and `@ExceptionHandler`.

```
java
@ControllerAdvice
public class GlobalExceptionHandler {
    @ExceptionHandler(Exception.class)
    public ResponseEntity<?> handle(Exception ex) {
        return ResponseEntity.status(500).body("Error: " + ex.getMessage());
```

```
    }
}
```

13. What is ResponseEntity?

It wraps the response body and status code. Used for custom HTTP responses.

14. How to validate request body in Spring Boot?

Use `@Valid` and JSR-303 annotations like `@NotNull`, `@Size`, etc.

```
java
@PostMapping("/users")
public ResponseEntity<?> save(@Valid @RequestBody User user) { ... }
```

15. How to enable CORS in Spring Boot?

```
java
@CrossOrigin(origins = "*")
```

🌐 PART 2: Microservices Concepts (16–30)

16. What is a Microservice?

An independent, small service that does one job and communicates over REST, often via JSON or Message Queues.

17. Advantages of Microservices

- Scalability
- Independent deployment
- Technology flexibility
- Faster development

18. Challenges of Microservices

- Complex debugging
- Network latency
- Distributed transactions
- Service coordination

19. What is Service Registry & Discovery?

Tools like **Eureka** let microservices find each other dynamically without hardcoding URLs.

20. What is Eureka?

A Netflix-based service registry for service discovery in Spring Cloud.

21. How do microservices communicate?

- REST APIs (via HTTP)
- Message Brokers (Kafka, RabbitMQ)
- Feign Clients (declarative REST client)

22. What is Feign Client?

It simplifies REST client code using interfaces.

```
java
@FeignClient(name = "user-service")
public interface UserClient {
    @GetMapping("/users/{id}")
    User getUser(@PathVariable String id);
}
```

23. What is API Gateway?

A single entry point for routing requests to microservices, handling:

- Authentication
- Rate Limiting
- Logging
- CORS

Example: **Spring Cloud Gateway**, **Zuul**

24. What is Circuit Breaker?

Protects microservices from failures using fallback logic. Tool: **Resilience4j**, **Hystrix**.

25. What is Load Balancing?

Distributes traffic between instances. Tool: Ribbon (client-side), Spring Cloud LoadBalancer.

26. What is Config Server in Spring Cloud?

Centralized configuration manager. Stores configs in Git and auto-refreshes microservices.

27. How do you secure microservices?

- OAuth2 / JWT tokens
- API Gateway with Auth filters
- Spring Security

28. What is Swagger?

Generates interactive REST API documentation with UI.

29. Difference between Monolith and Microservice

Monolith	Microservice
All-in-one	Small, focused units
Hard to scale	Scalable independently
Tight coupling	Loose coupling

30. What is the role of DTOs?

- Transfer data between layers
- Decouple entity from API response

🛠️ PART 3: Real-Time Scenarios & Advanced (31–50)

31. How to handle pagination in Spring Boot API?

Use Spring Data's `Pageable` interface.

```
java
@GetMapping("/users")
public Page<User> getAll(Pageable pageable) {
    return userRepo.findAll(pageable);
}
```

32. How to write custom query in JPA?

```
java
@Query("SELECT u FROM User u WHERE u.email = ?1")
User findByEmail(String email);
```

33. How do you test Spring Boot APIs?

- Use `@SpringBootTest`, `MockMvc`, and JUnit.
- Also use Postman or Swagger UI for manual testing.

34. How to log API requests/responses?

Use `OncePerRequestFilter` and log inside `doFilterInternal`.

35. How to schedule tasks in Spring Boot?

Use `@Scheduled` with `@EnableScheduling`.

```
java
@Scheduled(fixedRate = 5000)
public void runTask() {
    System.out.println("Runs every 5 sec");
}
```

36. How do you deploy Spring Boot apps?

- As a JAR using `java -jar app.jar`
- Dockerize and deploy to cloud/K8s

37. What is Actuator in Spring Boot?

Gives health, metrics, beans, env, etc. Use `spring-boot-starter-actuator`.

38. How do you handle JSON serialization issues?

- Use `@JsonIgnore`, `@JsonProperty`, etc. from Jackson.

39. What are profiles in Spring Boot?

Enable multiple environments (dev, prod, test) using `@Profile` and config files like:

- `application-dev.properties`
- `application-prod.properties`

40. What is H2 Database?

An in-memory DB used for testing/dev. Add dependency and configure via `application.properties`.

41–50: 🔥 *Bonus Real-Time Tips*

41. Always return meaningful HTTP status codes (`200 OK`, `404 Not Found`, `400 Bad Request`).
42. Use `@Value` or `@ConfigurationProperties` to read properties into beans.
43. Implement centralized logging using `Slf4j` and `Logback`.
44. Avoid using `Entity` directly in response – use DTO.
45. Know how to use Postman + Swagger to test endpoints.
46. Use `ModelMapper` or `MapStruct` for entity-DTO conversion.
47. Always validate inputs – avoid trusting frontend blindly.
48. Make services stateless – don't store session data.
49. Apply exception handling at controller and service levels.
50. Understand how to handle **caching** (Spring Cache, Redis) and **rate limiting** (Bucket4j).

💡 Want a summarized PDF or cheat sheet for this too?

I can format both:

- Top 100 Java Coding Qs ✅
- Top 50 Spring Boot + Microservices Qs ✅
  Into a clean, revision-friendly PDF or Notion page.

Want me to create it for you?
